

PRAKTIKUM SISTEM OPERASI

MODUL IX

SYSCALL



Oleh:

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2311104049

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

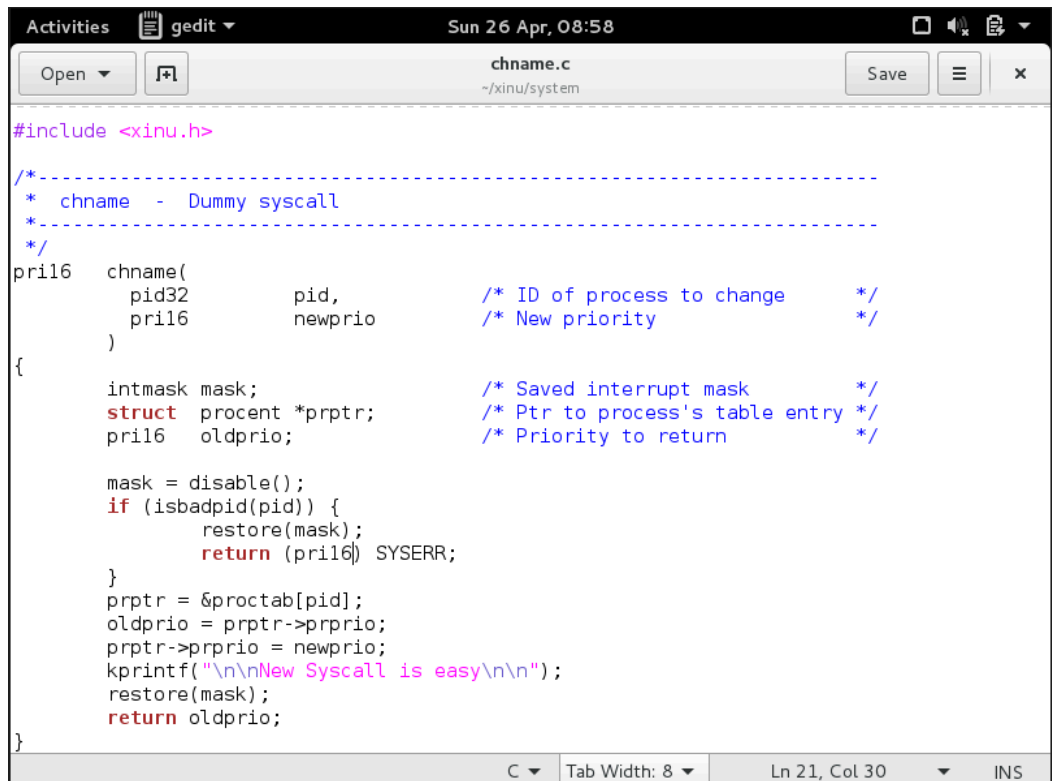
2026

JURNAL

1. Soal 1

Buat syscall baru seperti yang ditunjukkan pada modul syscall poin 9.5! (sertakan Screenshot kode dan hasil run)

- Syntax pada “cname.c” pada directory system

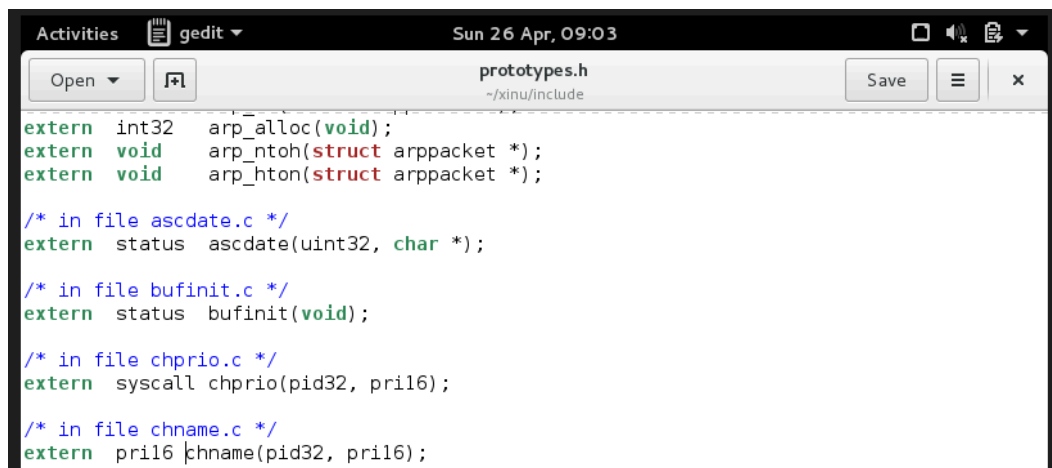


```
#include <xinu.h>

/*-----
 *  cname - Dummy syscall
 *-----
 */
pri16 cname(
    pid32    pid,          /* ID of process to change */
    pri16    newprio       /* New priority */
)
{
    intmask mask;          /* Saved interrupt mask */
    struct proctab *prptr; /* Ptr to process's table entry */
    pri16 oldprio;         /* Priority to return */

    mask = disable();
    if (isbadpid(pid)) {
        restore(mask);
        return (pri16) SYSERR;
    }
    prptr = &proctab[pid];
    oldprio = prptr->prprio;
    prptr->prprio = newprio;
    kprintf("\n\nNew Syscall is easy\n\n");
    restore(mask);
    return oldprio;
}
```

- Syntax pada “prototypes.h” pada directory include



```
extern int32 arp_alloc(void);
extern void arp_ntoh(struct arppacket *);
extern void arp_hton(struct arppacket *);

/* in file ascdatetime.c */
extern status ascdatetime(uint32, char *);

/* in file bufinit.c */
extern status bufinit(void);

/* in file chprio.c */
extern syscall chprio(pid32, pri16);

/* in file cname.c */
extern pri16 cname(pid32, pri16);
```

- Hasil running

```

Activities  Terminal  Sun 26 Apr, 09:08
xinu@xinu-develop-end: ~/xinu/include
File Edit View Search Terminal Help
[0x0011FEE0 to 0x001442C7]
Obtained IP address 10.0.3.15 (0x0a00030f)

-----2311104049-----

XINU

-----ZAENARIF PUTRA 'AINURDIN-----

Welcome to Xinu!

xsh $ uptime

New Syscall is easy

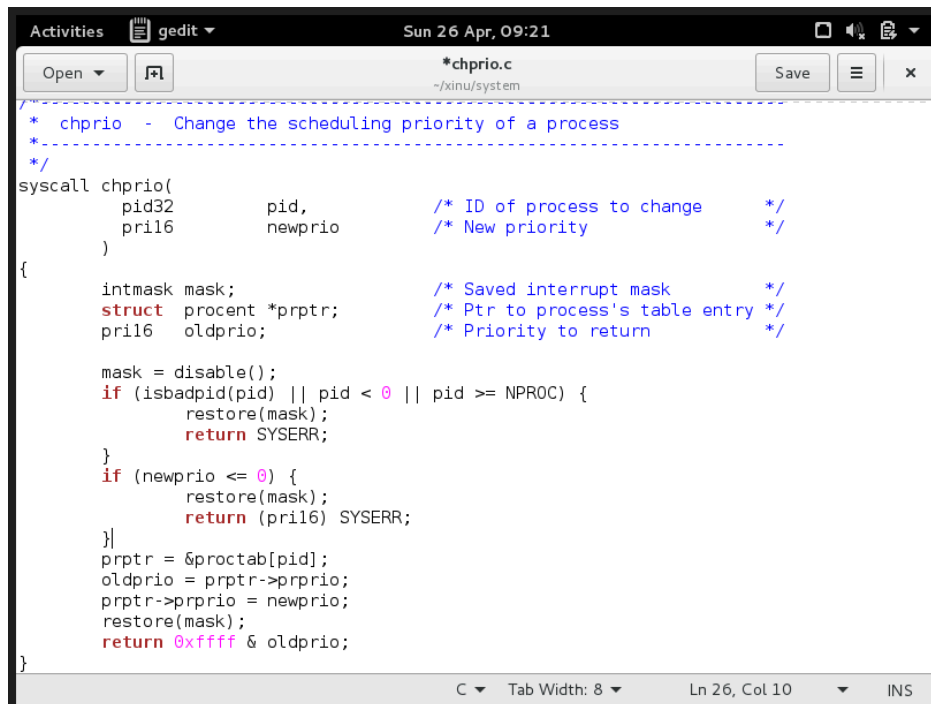
Xinu has been up 1 minute(s)
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:1 | ttyS0

```

2. Soal 2

Perbaiki syscall chprio (xinu/system/chprio.c) dengan memperhatikan validasi input

- Pastikan id adalah angka dari 0 – NPROC (ukuran maks banyaknya proses)
- Pastikan prioritas adalah bilangan yang positif
- Berikut Hasil Syntax yang sudah di update sesuai dengan permintaan

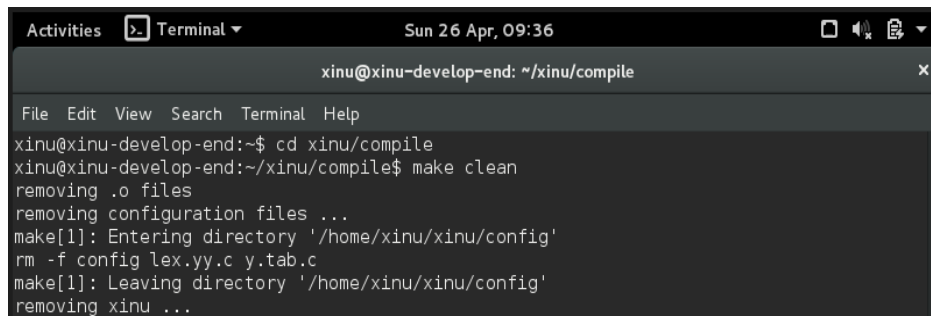


```
/* chprio - Change the scheduling priority of a process
 *
 */
syscall chprio(
    pid32    pid,      /* ID of process to change */
    pri16    newprio   /* New priority */
)
{
    intmask mask;      /* Saved interrupt mask */
    struct procent *prptr; /* Ptr to process's table entry */
    pri16    oldprio;   /* Priority to return */

    mask = disable();
    if (isbadpid(pid) || pid < 0 || pid >= NPROC) {
        restore(mask);
        return SYSERR;
    }
    if (newprio <= 0) {
        restore(mask);
        return (pri16) SYSERR;
    }
    prptr = &proctab[pid];
    oldprio = prptr->prprio;
    prptr->prprio = newprio;
    restore(mask);
    return 0xffff & oldprio;
}
```

Compile dan jalankan Xinu dengan syscall yang telah diperbaiki

- Make clean



```
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ cd xinu/compile
xinu@xinu-develop-end:~/xinu/compile$ make clean
removing .o files
removing configuration files ...
make[1]: Entering directory '/home/xinu/xinu/config'
rm -f config lex.yy.c y.tab.c
make[1]: Leaving directory '/home/xinu/xinu/config'
removing xinu ...
```

- Make

```

xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"\"cat version\"\" -I../i
nclude -o binaries/xsh_udpsrv.o ../shell/xsh_udpsrv.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"\"cat version\"\" -I../i
nclude -o binaries/xsh_uptime.o ../shell/xsh_uptime.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"\"cat version\"\" -I../i
nclude -o binaries/clkdsp.o ../system/clkdsp.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"\"cat version\"\" -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"\"cat version\"\" -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"\"cat version\"\" -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"\"cat version\"\" -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S

Loading object files to produce GRUB bootable xinu.
Copying xinu.elf to xinu.boot in the TFTP directory.

```

3. Soal 3

Lakukan hal-hal berikut ini :

- Edit xsh_uptime.c, Tambahkan kode berikut

```

/* Check for valid number of arguments */

if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

// tambahan code
chprio(5,33) // memanggil syscall chprio; mengubah prioritas menjadi 33 pada id=5

/* Check for valid number of arguments */

if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}
chprio(5, 33);

secs = clktime; /* Total seconds since boot */

```

- Compile source code tersebut dengan perintah: (Make & Make Clean)
 - Make clean

```

xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Tabs Help
xinu@xinu-develop-end: ~/xinu/compile xinu@xinu-develop-end: ~/xinu/shell
xinu@xinu-develop-end:~$ cd xinu/shell
xinu@xinu-develop-end:~/xinu/shell$ gedit xsh_uptime.c
xinu@xinu-develop-end:~/xinu/shell$ cd ..
xinu@xinu-develop-end:~/xinu$ cd compile
xinu@xinu-develop-end:~/xinu/compile$ make clean
removing .o files
removing configuration files ...
make[1]: Entering directory '/home/xinu/xinu/config'
rm -f config lex.yy.c y.tab.c
make[1]: Leaving directory '/home/xinu/xinu/config'
removing xinu ...

```

- Make

```

xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Tabs Help
xinu@xinu-develop-end: ~/xinu/compile xinu@xinu-develop-end: ~/xinu/shell
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"`cat version`\" -I../i
nclude -o binaries/xsh udpserver.o ../shell/xsh_udpserver.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"`cat version`\" -I../i
nclude -o binaries/xsh uptime.o ../shell/xsh_uptime.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"`cat version`\" -I../i
nclude -o binaries/clkdsp.o ../system/clkdsp.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"`cat version`\" -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"`cat version`\" -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"`cat version`\" -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"`cat version`\" -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S

Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.

xinu@xinu-develop-end:~/xinu/compile$

```

- Jalankan perintah ps

```

Activities Terminal Sun 26 Apr, 09:46
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

-----2311104049-----
XINU
-----ZAENARIF PUTRA 'AINURDIN-----

Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout       wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  50   4 0x005D7FFC 0x005D79AC 8192 0 0
6 ps         curr  20   5 0x005F7FFC 0x005F7FC4 8192 0 0
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0

```

- Jalankan uptime

```

Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout       wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  50   4 0x005D7FFC 0x005D79AC 8192 0 0
6 ps         curr  20   5 0x005F7FFC 0x005F7FC4 8192 0 0
xsh $ uptime
Xinu has been up 0

New Syscall is easy

minute(s)
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0

```

- Jalankan ps

```

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout       wait  20   0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  33   4 0x005D7FFC 0x005D79AC 8192 0 7
8 ps         curr  20   5 0x005F7FFC 0x005F7FC4 8192 0 0
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0

```

- Testing prioritas tidak boleh < 0 : Ubah “chprio(5,33)” menjadi “chprio(5,-3)” pada xsh_uptime.c

```

/* Check for valid number of arguments */
if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
               args[0]);
    return 1;
}
chprio(5, -3);

secs = clktime; /* Total seconds since boot */

/* Calculate total minutes */
mins = secs / secpermin;

/* Print the uptime strictly in minutes*/
printf("Xinu has been up %d minute(s)\n", mins);
chname(1,20);

return 0;
}

```

C Tab Width: 8 Ln 36, Col 20 INS

Activities Terminal Sun 26 Apr, 10:04

xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Tabs Help

xinu@xinu-develop-end: ~/xinu/compile xinu@xinu-develop-end: ~/xinu/shell

```

-----
0 prnull    ready    0      0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout     wait     500    0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin     wait     500    0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv     20    3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell     recv     50    4 0x005D7FFC 0x005D79AC 8192 0 0
6 ps        curr     20    5 0x005F7FFC 0x005F7FC4 8192 0 0
xsh $ uptime
Xinu has been up 0 minute(s)

New Syscall is easy

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout       wait   20    0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500   0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20    3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50    4 0x005D7FFC 0x005D79AC 8192 0 7
8 ps          curr   20    5 0x005F7FFC 0x005F7FC4 8192 0 0
xsh $

```


- Testing id adalah valid: Ubah “chprio(5,33)” menjadi “chprio(3000,3)”

```

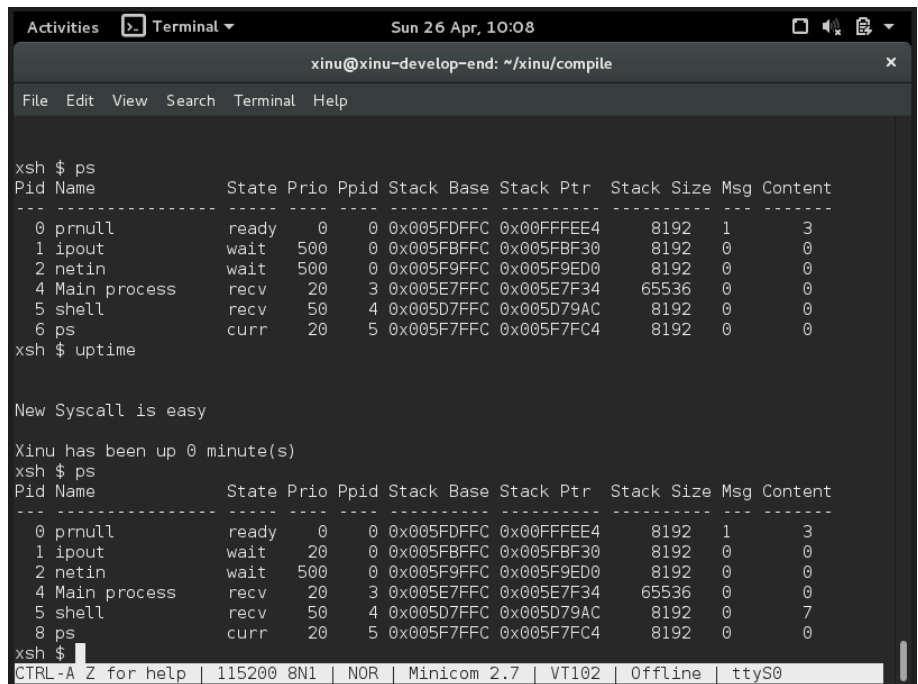
/* Check for valid number of arguments */

if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
               args[0]);
    return 1;
}
chprio(3000, 3);

secs = clktime;          /* Total seconds since boot */

/* Calculate total minutes */
mins = secs / secpermin;

```



Hasil dua testing di atas adalah prioritas tidak berubah karena salah argument (dibuktikan dengan menggunakan perintah ps)